Global Arbitrary-Order Enclosures for the SU(2) Heat-Kernel Action

Image remainders, a three-chart theorem, and validated rational bounds

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Abstract

We give explicit arbitrary-order derivative enclosures for the central heat kernel and its negative logarithm on SU(2), regarded as the unit three-sphere with heat generator $\frac{1}{2}\Delta$. Opposite Poisson images are paired before division by the Weyl denominator. On the identity chart this gives a sharp exponentially small correction to the Gaussian action. A second majorant, obtained by maximizing each exponential–polynomial factor in heat time, controls the transition chart without an order-dependent small-time restriction. At the cut locus we pair odd affine-Weyl images, prove an explicit positive principal-pair lower bound, and propagate differentiated image bounds through the logarithm. These estimates combine into a global three-chart theorem on the full class interval.

The antipode is treated as a Morse–Bott cut point: the two shortest radial image terms encode an S^2-family of minimizing geodesics and give $p_t(\pi) \asymp t^{-5/2}e^{-\pi^2/(2t)}$. Exact normalization also yields a Haar-probability tail bound. A rational validation record instantiates orders zero through eight at $t_0 = 1/5$ and records the positivity margins and all derivative orders using exact integer comparisons. In lattice normalization the estimates give explicit weak-field, convexity, and nonlinear-remainder bounds for a plaquette action. They do not establish an interacting-measure estimate, a renormalization-group trajectory, continuum Yang–Mills theory, confinement, or a mass gap.

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1 Introduction

The heat kernel on a compact Lie group is simultaneously a spectral object, a short-time Gaussian, and a convolution kernel. These descriptions expose different structures. Character sums make normalization and the semigroup law transparent; Poisson images expose the ultraviolet Gaussian; neither description by itself is a stable formula at every class angle and heat time. The negative logarithm compounds the problem: differentiating it requires a strict lower bound before quotient estimates become legitimate.

This issue is concrete in lattice gauge theory. The heat-kernel action of Menotti and Onofri is classical [MO81], as are its relation to the Manton action [Man80] and its exact convolution behavior in two-dimensional Yang–Mills theory [GKS89, Sen93]. Constructive and validated estimates require more specific data: a declared normalization, a closed interval, derivatives of $-\log p_t$ through a prescribed order, an explicit truncation rule, and a rigorously controlled positive lower bound.

The central result of this note is a global version of these estimates for SU(2). Given any finite order M, the theorem produces explicit bounds on three overlapping class-angle charts. The identity and transition charts use the same paired correction with different time majorants. The cut chart uses positive odd-image pairs and a principal-pair lower bound. The construction is not a new formula for the heat kernel. Its contribution is an explicit finite-time enclosure calculus, including the antipodal Morse–Bott prefactor and an exact-rational numerical realization.

The proof has four ingredients.

(i) Opposite images are paired before division by the class angle.

(ii) An integral identity for $\sinh u/u$ gives a derivative majorant with one expression valid at the identity and in the interior.

(iii) Exponential partial Bell polynomials transfer the hierarchy through the logarithm without repeating an order-by-order quotient calculation.

(iv) At the antipode, the two shortest radial image terms are paired before taking a removable limit; the resulting prefactor reflects the continuous family of minimizing geodesics.

The paper is organized from exact formulas to validated consequences. Sections 2–4 fix conventions and prove the derivative hierarchy. Sections 5 and 6 treat the action Hessian and endpoint charts. Section 7 proves normalized tails. Section 8 describes the rational validation, and Section 9 derives the lattice normalization and a concrete plaquette estimate. Complete summation, endpoint, and validation details are deferred to the appendices.

2 Geometry and normalization

We identify SU(2) with the unit sphere S^3. Every conjugacy class has a representative

$$g_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta}), \qquad 0 \le \theta \le \pi.$$

Thus $\frac{1}{2}\operatorname{Tr} g_\theta = \cos\theta$. The Riemannian distance from the identity is θ. Normalized Haar probability has radial part

$$\mathrm{d}H(\theta) = \frac{2}{\pi}\sin^2\theta \, \mathrm{d}\theta. \tag{1}$$

Our Laplace–Beltrami operator is nonpositive, and the heat semigroup is $e^{t\Delta/2}$. The representation indexed by $n \in \mathbb{N}_0$ has dimension $n+1$, character

$$\chi_n(\theta) = \frac{\sin((n+1)\theta)}{\sin\theta},$$

and $-\Delta$ eigenvalue $n(n+2)$. Thus the leading short-time action is $\theta^2/(2t)$.

Object	convention here	conversion warning
group metric	unit S^3	Casimir may be scaled by 2 or 4
heat equation	$\partial_t u = \frac{1}{2}\Delta u$	replace t under other generators
class angle	$[0, \pi]$	some physics sources use half the rotation angle
Haar measure	probability	volume measure has total mass $2\pi^2$
action	$s_t = -\log p_t$	additive constants do not affect derivatives

Table 1: Convention table. Every validated numerical bound uses the middle column exactly.

These choices are part of every numerical statement. In the common physics basis $T^a = \sigma^a/2$, with $\operatorname{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$, one has $g_\theta = \exp(2i\theta T^3)$. Consequently a Lie-algebra coordinate written as $\exp(iX^a T^a)$ has $|X| = 2\theta$ in this metric convention. Constants must be rescaled before comparison with a convention using $|X|$ itself as the geodesic distance.

3 Spectral and image representations

Proposition 3.1 (Character expansion)**.** *For $t > 0$ the central heat kernel with respect to Haar probability is*

$$p_t(\theta) = \sum_{n=0}^{\infty} (n+1) e^{-n(n+2)t/2} \frac{\sin((n+1)\theta)}{\sin\theta}, \tag{2}$$

with the endpoint values understood by continuity. The series and all its spatial derivatives converge normally for t bounded below by a positive number, and $\int p_t \, \mathrm{d}H = 1$.

Proof. Peter–Weyl expansion of the delta mass supplies the coefficient $n+1$; the declared generator supplies the exponential eigenvalue. Character orthogonality leaves only the trivial representation after Haar integration. □

Proposition 3.2 (Poisson-image formula)**.** *For $0 < \theta < \pi$,*

$$p_t(\theta) = C_t \frac{1}{\sin\theta} \sum_{k\in\mathbb{Z}} (\theta + 2\pi k) e^{-(\theta+2\pi k)^2/(2t)}, \qquad C_t = \frac{e^{t/2}\sqrt{2\pi}}{2t^{3/2}}. \tag{3}$$

Both endpoints have removable limits after the appropriate images are paired. Equations (2) *and* (3) *agree exactly for every $t > 0$.*

The proof by differentiated Poisson summation is given in Appendix A. Formula (2) is stable for moderate and large time. Formula (3) is stable at small time away from the antipode. Near the antipode one must first pair the two shortest images.

For $a_k = 2\pi k$, factor the zero image in (3):

$$p_t(\theta) = C_t e^{-\theta^2/(2t)} \frac{\theta}{\sin\theta}(1 + q_t(\theta)), \tag{4}$$

where the paired correction is

$$q_t(\theta) = 2\sum_{k\geq 1} e^{-a_k^2/(2t)} \left[\cosh\frac{a_k\theta}{t} - \frac{a_k}{\theta}\sinh\frac{a_k\theta}{t}\right]. \tag{5}$$

At $\theta = 0$ the quotient is interpreted by its even Taylor series.

4 An arbitrary-order image hierarchy

The identity

$$\frac{a}{\theta}\sinh\frac{a\theta}{t} = \frac{a^2}{t}\int_0^1 \cosh\frac{sa\theta}{t}\,\mathrm{d}s \tag{6}$$

removes the apparent singularity in (5) before differentiation.

Fix $M \in \mathbb{N}_0$, $0 < \rho < \pi$, and $t_0 > 0$. For $0 \leq m \leq M$ define

$$Q_m(t_0,\rho) = 2\sum_{k\geq 1} e^{-a_k^2/(2t_0)+a_k\rho/t_0} \left(\frac{a_k}{t_0}\right)^m \left(1 + \frac{a_k^2}{t_0(m+1)}\right). \tag{7}$$

The series is explicit and computable. If it is stopped after K terms, the omitted tail is at most the first omitted term divided by $1 - r_{m,K+1}$, where

$$r_{m,N} = \left(\frac{N+1}{N}\right)^{m+2} \exp\left[-\frac{2\pi^2(2N+1) - 2\pi\rho}{t_0}\right]. \tag{8}$$

Theorem 4.1 (Uniform arbitrary-order image enclosure)**.** *Suppose $0 < \rho < \pi$ and*

$$t_0 \leq \frac{(2\pi)^2/2 - 2\pi\rho}{M+1}. \tag{9}$$

Then, for every $0 < t \leq t_0$, $0 \leq \theta \leq \rho$, and $0 \leq m \leq M$,

$$\left|q_t^{(m)}(\theta)\right| \leq Q_m(t_0,\rho). \tag{10}$$

For each fixed M, t_0, ρ, the right-hand side has a rigorously bounded geometric tail of the form (8)*.*

Proof. Differentiate (6) under the integral. Each hyperbolic derivative is bounded by $e^{a\rho/t}$, while the integral contributes $\int_0^1 s^m \, \mathrm{d}s = (m+1)^{-1}$. This gives the kth summand in (7). Writing its t-dependence as $e^{-A_k/t} t^{-m}(1 + c_k/t)$ shows that it increases with t whenever $t \le A_k/(m+1)$. The smallest $A_k = a_k^2/2 - a_k\rho$ occurs at $k = 1$; (9) therefore makes $t = t_0$ the uniform worst case. Finally, the quotient of successive summands is bounded by (8); details appear in Appendix B. □

The theorem is arbitrary-order in the standard effective sense: for every requested finite M it gives a closed computation. It is not a claim that a single fixed t_0 satisfies (9) for all M at once.

5 Logarithmic derivatives and the action Hessian

Assume $Q_0 < 1$ and let $\mathsf{B}_{m,\ell}$ denote the exponential partial Bell polynomial. Define

$$B_m = \sum_{\ell=1}^{m} \frac{(\ell-1)!}{(1-Q_0)^\ell} \mathsf{B}_{m,\ell}(Q_1, \dots, Q_{m-\ell+1}). \tag{11}$$

Also set

$$G_m(\rho) = \sum_{n \ge \lceil m/2 \rceil} \frac{\zeta(2n)}{n\pi^{2n}} \frac{(2n)!}{(2n-m)!} \rho^{2n-m}. \tag{12}$$

Theorem 5.1 (Action derivative enclosure)**.** *Under the hypotheses of Theorem 4.1, if $Q_0 < 1$, then for $1 \le m \le M$,*

$$\sup_{0<t\le t_0} \sup_{0\le\theta\le\rho} \left| \partial_\theta^m \left(s_t(\theta) - \frac{\theta^2}{2t} \right) \right| \le G_m(\rho) + B_m, \qquad s_t = -\log p_t. \tag{13}$$

In particular $1 + q_t \ge 1 - Q_0 > 0$, so every logarithmic differentiation is justified by an explicit positivity floor.

Proof. Taking the logarithm of (4) gives

$$s_t(\theta) = \frac{\theta^2}{2t} + \log \frac{\sin\theta}{\theta} - \log(1 + q_t(\theta)) - \log C_t. \tag{14}$$

Euler's product yields the absolutely convergent expansion

$$\log \frac{\sin\theta}{\theta} = -\sum_{n\ge1} \frac{\zeta(2n)}{n\pi^{2n}} \theta^{2n}, \qquad |\theta| < \pi,$$

and hence (12). Faà di Bruno's formula for $\log(1 + q_t)$ gives (11) after absolute values. See Appendices C and D. □

For a radial function $f(\theta)$ on the unit S^3, the Hessian eigenvalues are

$$\lambda_{\mathrm{rad}} = f''(\theta), \qquad \lambda_{\tan} = f'(\theta)\cot\theta \quad \text{(multiplicity two)}. \tag{15}$$

Corollary 5.2 (Local convexity budget)**.** *Let $R_2 = G_2 + B_2$ and $0 < \rho < \pi/2$. On $0 \le \theta \le \rho$,*

$$\lambda_{\mathrm{rad}}(s_t) \ge t^{-1} - R_2, \tag{16}$$

$$\lambda_{\tan}(s_t) \ge \rho \cot\rho \, t^{-1} - R_2. \tag{17}$$

Thus the action is strictly geodesically convex wherever the displayed lower bounds are positive.

Proof. The radial inequality is immediate. If $r = s_t - \theta^2/(2t)$, evenness gives $r'(0) = 0$, hence $|r'(\theta)| \le R_2\theta$. Since $0 \le \theta\cot\theta \le 1$ and it decreases on $[0, \rho]$, substitution in (15) proves the tangential bound. □

6 A global three-chart enclosure atlas

No single numerical representation is well conditioned across the whole class interval. Choose $0 < \rho < \pi/2$ and $0 < \delta < \pi/2$. We use

$$[0, \rho] \quad \text{(identity chart)}, \qquad [\rho, \pi - \delta] \quad \text{(transition chart)}, \qquad [\pi - \delta, \pi] \quad \text{(cut chart)}.$$

The overlaps allow the spectral and image formulas to be compared before either representation loses numerical stability.

Theorem 4.1 gives the sharp identity-chart constants. For the larger interval $[0, R]$, $R < \pi$, put

$$A_k(R) = \frac{a_k^2}{2} - a_k R, qquada_k = 2\pi k,$$

and, for $A > 0$, define

$$\mathcal{H}_0(A, t_0) = e^{-A/t_0}, \qquad \mathcal{H}_r(A, t_0) = e^{-A/\tau_r} \tau_r^{-r}, \quad \tau_r = \min\{t_0, A/r\} \quad (r \geq 1). \tag{18}$$

This is the exact maximum of $e^{-A/t} t^{-r}$ over $0 < t \leq t_0$. Set

$$\widehat{Q}_m(t_0, R) = 2 \sum_{k \geq 1} a_k^m \left[\mathcal{H}_m(A_k(R), t_0) + \frac{a_k^2}{m+1} \mathcal{H}_{m+1}(A_k(R), t_0) \right]. \tag{19}$$

Unlike (7), this majorant needs no order-dependent monotonicity condition. Its tail is Gaussian because $\tau_r = t_0$ for all sufficiently large k. Let $\widehat{B}_m$ be the expression in (11) with Q_j replaced by $\widehat{Q}_j$.

At the cut locus put $\phi = \pi - \theta$ and pair images equidistant from odd multiples $b_j = (2j+1)\pi$, $j \in \mathbb{N}_0$.

Proposition 6.1 (Positive cut-locus pairs)**.** *The jth paired numerator in* (3) *is*

$$2e^{-(b_j^2 + \phi^2)/(2t)} \left[b_j \sinh \frac{b_j \phi}{t} - \phi \cosh \frac{b_j \phi}{t} \right]. \tag{20}$$

After division by $\sin\phi$ it has a removable positive limit at $\phi = 0$ whenever $b_j^2 > t$. It is positive throughout $0 \leq \phi \leq \delta$ whenever

$$b_j^2 > t + b_j \delta. \tag{21}$$

In particular all pairs are positive if $\pi^2 > t + \pi\delta$.

The proof appears in Appendix E. The paired terms are radial affine-Weyl images. They must not be identified with the minimizing geodesics: between the identity and the antipode of S^3 there is an S^2-family of minimizing geodesics.

Appendix F defines an explicit principal-pair lower bound $L_{\mathrm{cut}}(t, \delta)$, derivative majorants $D_m^{\mathrm{cut}}(t, \delta)$ for p_t, and the associated Bell-polynomial bounds $C_m^{\mathrm{cut}}(t, \delta)$ for $-\log p_t$.

Theorem 6.2 (Global three-chart derivative enclosure)**.** *Fix $M \in \mathbb{N}$, $0 < \rho < \pi/2$, $0 < \delta < \pi/2$, and $t_0 > 0$. Assume the hypotheses of Theorem 4.1 on $[0, \rho]$,*

$$Q_0(t_0, \rho) < 1, \qquad \widehat{Q}_0(t_0, \pi - \delta) < 1, \qquad \pi^2 > t_0 + \pi\delta.$$

Then, for every $0 < t \le t_0$ *and* $1 \le m \le M$,

$$\sup_{0\le\theta\le\rho}\left|\partial_\theta^m\left(s_t-\frac{\theta^2}{2t}\right)\right| \le G_m(\rho)+B_m, \tag{22}$$

$$\sup_{\rho\le\theta\le\pi-\delta}\left|\partial_\theta^m\left(s_t-\frac{\theta^2}{2t}\right)\right| \le G_m(\pi-\delta)+\widehat{B}_m, \tag{23}$$

$$\sup_{\pi-\delta\le\theta\le\pi}|\partial_\theta^m s_t(\theta)| \le C_m^{\mathrm{cut}}(t,\delta). \tag{24}$$

All quantities on the right are convergent series with explicit geometric or Gaussian tails. Thus the three estimates form a computable cover of the full class interval $[0,\pi]$.

The cut-chart bound depends on t and cannot be made uniform as $t \downarrow 0$. This is intrinsic: logarithmic derivatives detect the cut locus and the Hessian has stronger small-time growth there.

Theorem 6.3 (Explicit small-time forms)**.** *On the identity chart, for* $1 \le m \le M$,

$$\left|\partial_\theta^m\left(ts_t(\theta)-\frac{\theta^2}{2}\right)-t\,\partial_\theta^m\log\frac{\sin\theta}{\theta}\right| \le tB_m. \tag{25}$$

At the antipode,

$$p_t(\pi)=e^{t/2}\sqrt{2\pi}\,\pi^2t^{-5/2}e^{-\pi^2/(2t)}\left[1-\frac{t}{\pi^2}+O\Big(e^{-4\pi^2/t}\Big)\right], \tag{26}$$

and

$$s_t''(\pi)=-\frac{\pi^2}{3t^2}+O(t^{-1}). \tag{27}$$

The power $t^{-5/2}$ in (26), rather than the generic off-cut-locus $t^{-3/2}$ power in three dimensions, is the Morse–Bott contribution of the two-dimensional family of minimizing geodesics. Equation (27) also shows why the local convexity result of Corollary 5.2 cannot extend globally.

7 Haar-normalized tails

Normalization need not be estimated numerically: Proposition 3.1 gives $\int p_t\,\mathrm{d}H = 1$ exactly. Let $\Pr_t$ denote the probability measure on SU(2) with density p_t relative to normalized Haar measure. Combining (3) with (1) cancels one sine denominator.

Theorem 7.1 (Explicit normalized tail)**.** *Let* $0 < \rho < \pi$ *and* $t > 0$. *Define*

$$E(t)=t\frac{e^{-\pi^2/(2t)}+e^{-2\pi^2/t}}{1-e^{-4\pi^2/t}}. \tag{28}$$

Then

$$\Pr_t\{\theta\ge\rho\} \le e^{t/2}\sqrt{\frac{2}{\pi}}\,t^{-3/2}\left[t\left(\rho+\frac{t}{\rho}\right)e^{-\rho^2/(2t)}+E(t)\right]. \tag{29}$$

The right-hand side may be replaced by its minimum with one.

Proof. The zero-image radial numerator is bounded using $\sin\theta \le \theta$ and extension to infinity:

$$\int_\rho^\pi \theta\sin\theta e^{-\theta^2/(2t)}\,\mathrm{d}\theta \le \int_\rho^\infty \theta^2e^{-\theta^2/(2t)}\,\mathrm{d}\theta \le t\left(\rho+\frac{t}{\rho}\right)e^{-\rho^2/(2t)}.$$

The last inequality is the elementary Mills-ratio bound. All nonzero images contribute at most $E(t)$ in absolute value. Multiplication by $(2/\pi)C_t = e^{t/2}\sqrt{2/\pi}\,t^{-3/2}$ and exact normalization finish the proof. Appendix G gives the image integration details. □

At fixed ρ, (29) has logarithmic rate $-\rho^2/(2t)$ as $t \downarrow 0$; the nonzero images are exponentially smaller when $\rho < \pi$.

8 Validated rational bounds

The accompanying exact-rational computation instantiates the identity chart at

$$t_0 = \frac{1}{5}, \qquad \rho = 1, \qquad M = 8,$$

and the transition/cut overlap at $\delta = 1/5$. Machin's identity encloses π; alternating or positive Taylor series and compact exponential majorants bound elementary functions; rational square-root intervals are checked by squaring their endpoints. Every theorem-critical comparison is an integer comparison. Binary floating point is used only in the separate formula diagnostic.

m	Q_m upper	B_m upper	G_m upper
0	3.0×10^{-18}	—	—
1	4.3×10^{-17}	4.3×10^{-17}	0.371
2	8.94×10^{-16}	8.94×10^{-16}	0.455
3	2.12×10^{-14}	2.12×10^{-14}	0.289
4	5.35×10^{-13}	5.35×10^{-13}	0.503
5	1.41×10^{-11}	1.41×10^{-11}	0.845
6	3.81×10^{-10}	3.81×10^{-10}	2.086
7	1.06×10^{-8}	1.06×10^{-8}	5.677
8	2.95×10^{-7}	2.95×10^{-7}	18.84

Table 2: Upward-rounded decimal displays of the exact rational identity-chart bounds. The rational values are stored in the validation record.

The identity-chart positivity floor is greater than $1 - 3.0 \times 10^{-18}$. On the larger interval $0 \le \theta \le \pi - 1/5$, the time-maximized transition bound gives

$$\widehat{Q}_0 < 0.985, \qquad 1 - \widehat{Q}_0 > 0.0159.$$

The exact cut-pair condition has positive margin $\pi^2 - t_0 - \pi\delta > 9.03$. Thus every denominator used in the global three-chart theorem has a recorded positive lower bound. The transition derivative constants are intentionally conservative; all orders zero through eight, together with the transition-chart geometric constants, are included in the machine-readable record.

The normalized tail upper bounds at $\rho = 1$ are 0.595, 0.0913, and 0.00376 for $t = 1/5, 1/10, 1/20$. The validated plaquette margins are

$$\lambda_{\mathrm{rad}} > 4.545, \qquad \lambda_{\tan} > 2.755, \qquad tR_2 < 0.091, \quad tR_3 < 0.058, \quad tR_4 < 0.101.$$

The checker recomputes the file digest, rational ordering, both positivity floors, every Bell-polynomial logarithmic bound, the cut-pair margin, the plaquette bounds, and all geometric-tail contractions. The digest checks byte-level integrity only; it is not treated as a mathematical argument. A separate numerical diagnostic compares (2) and (3) on 21 cells, including both removable endpoints. Its maximum relative-or-absolute discrepancy is 1.91×10^{-15}. This diagnostic detects convention or implementation errors but is not used to prove an inequality.

9 Lattice normalization and a validated plaquette example

Let $T^a = \sigma^a/2$ and write a plaquette holonomy as

$$U_P = \exp(iX_P^a T^a), \qquad \frac{1}{2}\operatorname{Tr} U_P = \cos\theta_P, \qquad \theta_P = \frac{|X_P|}{2} \quad (|X_P| \le 2\pi).$$

The one-plaquette heat-kernel action is $s_t(\theta_P)$. In spin notation $n = 2j$, so the character weight in (2) is $e^{-2t\,j(j+1)}$. A convention that writes the weight as $e^{-\tau C_2(j)}$ therefore has $\tau = 2t$.

For a smooth continuum field,

$$X_P^a = a^2 g_0 F_{\mu\nu}^a(x) + O(a^3),$$

in the convention where g_0 appears in the plaquette exponential. Hence

$$s_t(U_P) = \frac{a^4 g_0^2}{8t} F_{\mu\nu}^a F_{\mu\nu}^a + O\left(\frac{a^5}{t}\right) + O(a^4 g_0^2 |F|^2), \tag{30}$$

up to the field-independent term $-\log C_t$. The final matching of t to the conventional bare coupling depends on whether g_0 is included in $F_{\mu\nu}$ and on the normalization of the continuum action; equation (30) makes that conversion explicit rather than silently identifying t with a standard β.

Corollary 9.1 (Local nonlinear budget)**.** *Fix $M \ge 3$ and an identity chart satisfying Theorem 5.1. Then*

$$\left|\partial_\theta^m \left(t s_t - \frac{\theta^2}{2}\right)\right| \le t(G_m + B_m), \qquad 1 \le m \le M. \tag{31}$$

For a one-parameter link variation with class angle $\theta_P(x)$,

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} s_t(\theta_P(x)) = s_t''(\theta_P)(\theta_P')^2 + s_t'(\theta_P)\theta_P'', \tag{32}$$

$$\left|\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(s_t - \frac{\theta_P^2}{2t}\right)\right| \le R_2(\theta_P')^2 + R_2|\theta_P|\,|\theta_P''|, \tag{33}$$

as long as the path remains in the chart. Radial plaquette variations have $\theta_P' = 1$ and $\theta_P'' = 0$.

For the rationally validated instance $t_0 = 1/5$ and $\rho = 1$, the record gives $R_2 = G_2 + B_2 < 0.455$. Therefore every $0 < t \le 1/5$ satisfies

$$\lambda_{\mathrm{rad}}(s_t) > 4.545, \qquad \lambda_{\tan}(s_t) > 2.75 \quad \text{on } 0 \le \theta \le 1. \tag{34}$$

At $t = 1/5$, the scaled-action derivative corrections obey

$$tR_2 < 0.091, \qquad t(G_3 + B_3) < 0.058, \qquad t(G_4 + B_4) < 0.101.$$

Thus a commuting radial plaquette path $U_P(x) = \exp(2i(\theta + x)T^3)$ has a validated second derivative at least 4.545, while its third and fourth scaled derivatives are bounded by the last two displayed constants. This is a concrete finite-plaquette coercivity and nonlinearity statement, not a statement about a physical mass.

For a full lattice, composition with link coordinates requires bounds on the first and higher derivatives of the noncommutative plaquette product, and summation over plaquettes requires a specified operator norm. Gauge directions also prevent one-plaquette convexity from becoming a global strict-convexity statement on link space without gauge reduction.

Likewise, Theorem 7.1 concerns a random group element with density p_t relative to normalized Haar measure. In a four-dimensional lattice measure, plaquettes share links and obey nonabelian constraints. The theorem does not determine the interacting plaquette marginal or provide the large-field estimate needed for a renormalization-group argument.

10 Relation to existing estimates

Small-time heat-kernel asymptotics originate in the parametrix tradition [MP49] and Varadhan's logarithmic asymptotics [Var67]. Li–Yau estimates control positive heat-equation solutions geometrically [LY86]. These results apply far beyond the present model, but do not normally produce the convention-specific, finite-time array of numerical constants needed here.

The literature on logarithmic derivatives is closer to the present theorem. Chen, Li, and Wu obtain short-time estimates for the gradient and Hessian of the logarithmic heat kernel on complete Riemannian manifolds without curvature restrictions [CLW23]. Neel and Sacchelli treat arbitrary higher-order logarithmic derivatives, uniform localized expansions, and Morse–Bott cut loci, including standard Riemannian spheres [NS25]. Those works determine general asymptotic structure. The result here is different in logical form: for one compact group it gives explicit finite-t majorants, computable image-tail losses, positive lower bounds on each chart, and exact-rational numerical instances. The Morse–Bott interpretation of the antipode is consistent with the general cut-locus theory.

On compact Lie groups, representation theory gives the character expansion, and heat-kernel measures underlie the compact-group Segal–Bargmann transform [Hal94]. Explicit SU(2) formulas also appear in analyses of subelliptic heat kernels [BB09, ABGR09], whose operator and geometry differ from the Riemannian Laplacian used here.

In lattice gauge theory, the heat-kernel action of Menotti and Onofri [MO81] followed Manton's geometric action [Man80]. Exact convolution identities are central in two-dimensional Yang–Mills theory [GKS89, Sen93]. Heat-kernel actions have also appeared in studies of renormalization flow [BDJ85] and in modern work on real-time lattice actions [KW21]. The present paper does not establish a flow or a continuum limit. Its lattice contribution is a local analytic input: explicit control of the one-plaquette action and a precise account of what remains when plaquettes are coupled through shared links.

The contribution is therefore the combination of explicit constants, reproducible validation, and a complete three-chart specialization to SU(2). No priority claim is made for the heat-kernel formula, the general existence of logarithmic derivative asymptotics, or the Morse–Bott description of spherical cut loci.

11 Discussion and limitations

The main conceptual lesson is representational. Spectral normalization, local Gaussian structure, logarithmic differentiation, and cut-locus multiplicity are easiest in different coordinates. A rigorous calculation therefore needs a domain atlas and explicit transfer conditions rather than a single globally preferred formula.

The constants in Table 2 are intentionally conservative. Replacing the dyadic bound $e^{-x} \le 2^{-\lfloor x \rfloor}$ with rational range-reduced Taylor bounds improves them without changing the theorem. A branch-and-bound interval implementation could also subdivide t and θ to reduce dependency. Sharpness is secondary here to a short collection of elementary inequalities.

The global theorem separates two genuinely different regimes. After removal of the Gaussian action, the identity-chart image correction is exponentially small uniformly through any requested finite order. At the antipode the S^2-family of minimizing geodesics changes the heat-kernel prefactor and forces stronger growth of logarithmic derivatives. The cut-chart constants therefore retain their t dependence. This is a geometric feature, not a loss introduced by the proof.

Several extensions are natural but not proved. The Weyl denominator and affine Weyl group

suggest analogues for higher-rank compact groups, but walls intersect and the endpoint atlas becomes multidimensional. A result for $\mathrm{SU}(N)$ would require controlling those intersections, not simply replacing $2\pi k$ by a vector. Likewise, the local action budgets become an RG input only after dependent-measure, large-field, gauge, and topology estimates are supplied.

12 Conclusion

We constructed a global arbitrary-order enclosure calculus for the SU(2) heat-kernel action. Pairing images before differentiation makes the identity regular; time-maximized image bounds control the transition chart; positive odd-image pairs and a principal-pair lower bound control the cut chart. The antipodal formula displays the $t^{-5/2}$ Morse–Bott prefactor and the associated loss of convexity. Exact normalization yields a direct Haar-tail theorem, while the order-eight rational validation supplies reproducible numerical instances of the analytic bounds.

The result closes a finite representation problem: spectral and image descriptions now exchange normalization, derivative information, and endpoint geometry with explicit losses on the full class interval. For lattice gauge theory it provides a sharp one-plaquette small-field input. Transporting that local rigidity through a dependent, gauge-invariant multiscale measure remains a separate problem.

A Differentiated Poisson summation

Put $m = n + 1$ in (2). Since $n(n+2) = m^2 - 1$,

$$p_t(\theta) = \frac{e^{t/2}}{\sin\theta} \sum_{m\geq 1} m e^{-m^2 t/2} \sin(m\theta). \tag{35}$$

The Jacobi identity in the Fourier convention used here is

$$\sum_{m\in\mathbb{Z}} e^{-m^2 t/2} e^{im\theta} = \sqrt{\frac{2\pi}{t}} \sum_{k\in\mathbb{Z}} e^{-(\theta+2\pi k)^2/(2t)}.$$

Normal convergence permits differentiation in θ. Multiplying the derivative by -1 and taking real quantities gives

$$2\sum_{m\geq 1} m e^{-m^2 t/2} \sin(m\theta) = \sqrt{\frac{2\pi}{t}} \frac{1}{t} \sum_{k\in\mathbb{Z}} (\theta + 2\pi k) e^{-(\theta+2\pi k)^2/(2t)}.$$

Substitution in (35) yields (3), including the factor $C_t = e^{t/2}\sqrt{2\pi}/(2t^{3/2})$.

At $\theta = 0$, pair k and $-k$ before taking the limit. At $\theta = \pi$, pair the images whose arguments tend to opposite odd multiples of π. These operations are justified termwise by Gaussian domination. Thus the apparent zeros of the Weyl denominator are removable in the complete sum, but not necessarily in an unpaired truncation.

B Differentiated image tails

For $a > 0$, define

$$F_{a,t}(\theta) = e^{-a^2/(2t)} \left[\cosh\frac{a\theta}{t} - \frac{a^2}{t} \int_0^1 \cosh\frac{sa\theta}{t}\, \mathrm{d}s \right].$$

The hyperbolic function in the mth derivative is either a sine-hyperbolic or cosine-hyperbolic function. On $0 \le \theta \le \rho$ its absolute value is at most $e^{a\rho/t}$, and therefore

$$|\partial_\theta^m F_{a,t}(\theta)| \le e^{-a^2/(2t)+a\rho/t} \left(\frac{a}{t}\right)^m \left(1 + \frac{a^2}{t(m+1)}\right). \tag{36}$$

The integrand and all derivatives have a summable Gaussian majorant, so derivative interchange is valid at every finite order.

For $A = a^2/2 - a\rho > 0$, the logarithmic derivative with respect to t of $e^{-A/t}t^{-m}(1 + c/t)$ is bounded below by

$$\frac{A}{t^2} - \frac{m+1}{t}.$$

It is nonnegative for $t \le A/(m+1)$. This proves the time-monotonicity used in Theorem 4.1.

Let $T_{m,k}$ be twice the right side of (36) at $a = 2\pi k$ and $t = t_0$. Since

$$\frac{1 + c(k+1)^2}{1 + ck^2} \le \left(\frac{k+1}{k}\right)^2,$$

we obtain

$$\frac{T_{m,k+1}}{T_{m,k}} \le \left(\frac{k+1}{k}\right)^{m+2} e^{-[2\pi^2(2k+1)-2\pi\rho]/t_0}.$$

The right side decreases with k. If N is the first omitted index and $r_{m,N} < 1$, summing the geometric majorant gives

$$\sum_{k \ge N} T_{m,k} \le \frac{T_{m,N}}{1 - r_{m,N}}.$$

C Bell-polynomial logarithmic calculus

The exponential partial Bell polynomials satisfy

$$\mathsf{B}_{n,k}(x_1, \ldots) = \sum_{j=1}^{n-k+1} \binom{n-1}{j-1} x_j \mathsf{B}_{n-j,k-1}, \quad \mathsf{B}_{0,0} = 1. \tag{37}$$

Faà di Bruno's formula with $f(u) = \log u$ and $u = 1 + q$ gives

$$\partial^m \log(1+q) = \sum_{\ell=1}^{m} \frac{(-1)^{\ell-1}(\ell-1)!}{(1+q)^\ell} \mathsf{B}_{m,\ell}(q', \ldots, q^{(m-\ell+1)}).$$

The Bell polynomials have nonnegative integer coefficients. If $|q^{(j)}| \le Q_j$ and $|q| \le Q_0 < 1$, taking absolute values produces (11). The first three instances are

$$\begin{aligned} B_1 &= \frac{Q_1}{d}, \\ B_2 &= \frac{Q_2}{d} + \frac{Q_1^2}{d^2}, \\ B_3 &= \frac{Q_3}{d} + \frac{3Q_1Q_2}{d^2} + \frac{2Q_1^3}{d^3}, \qquad d = 1 - Q_0. \end{aligned}$$

Thus the familiar low-order quotient bounds are consequences of one general recurrence rather than separate calculations.

D The geometric sine correction

Euler's product on $|\theta| < \pi$ gives

$$\frac{\sin\theta}{\theta} = \prod_{j\geq 1}\left(1 - \frac{\theta^2}{\pi^2 j^2}\right).$$

Expanding each logarithm and interchanging absolutely convergent sums yields

$$\log\frac{\sin\theta}{\theta} = -\sum_{n\geq 1}\frac{\zeta(2n)}{n\pi^{2n}}\theta^{2n}.$$

Termwise differentiation proves (12). For exact-rational certification we use $\zeta(2n) \leq \zeta(2) = \pi^2/6$ and a rational lower bound for π. After a finite prefix, the factorial-polynomial ratio times $(\rho/\pi)^2$ provides a contractive geometric tail.

For reference, the exact low-order suprema on $[0, \rho]$, $\rho < \pi/2$, are bounded by

$$\sup\left|\partial_\theta^2 \log(\sin\theta/\theta)\right| \leq \csc^2\rho - \rho^{-2},$$
$$\sup\left|\partial_\theta^3 \log(\sin\theta/\theta)\right| \leq 2\rho^{-3} - 2\cos\rho\,\csc^3\rho.$$

The power-series majorant is less sharp but extends without a new symbolic derivation to every order.

E Antipodal pairing

Write $\theta = \pi - \phi$. The arguments of the two shortest radial images are $\pi - \phi$ and $-\pi - \phi$. Their numerator is

$$\begin{aligned}&(\pi-\phi)e^{-(\pi-\phi)^2/(2t)} - (\pi+\phi)e^{-(\pi+\phi)^2/(2t)}\\ &\quad = 2e^{-(\pi^2+\phi^2)/(2t)}\left[\pi\sinh\frac{\pi\phi}{t} - \phi\cosh\frac{\pi\phi}{t}\right].\end{aligned}$$

The same computation gives (20) for every $b_j = (2j+1)\pi$, $j \in \mathbb{N}_0$.

For $x \geq 0$, comparison of derivatives proves $\tanh x \geq x/(1+x)$. Hence, for $\phi > 0$,

$$b\tanh\frac{b\phi}{t} \geq \frac{b^2\phi}{t + b\phi} > \phi$$

under condition $b^2 > t + b\phi$. The bracket in (20) is therefore positive. Dividing its Taylor expansion by $\sin\phi$ gives the limit $b^2/t - 1$, times the positive common exponential factor. All later odd pairs satisfy a stronger inequality than the principal pair.

These are two terms of the affine-Weyl image sum, not two minimizing geodesics. At antipodal points of S^3, the initial tangent direction may be any point of the unit S^2, so the minimizing geodesics form a two-dimensional Morse–Bott family. The exact limit and its differentiated majorants are developed in Appendix F.

F Proof of the global atlas bounds

Transition chart

For $A > 0$ and $r \in \mathbb{N}_0$, elementary differentiation shows

$$\sup_{0<t\le t_0} e^{-A/t} t^{-r} = \mathcal{H}_r(A, t_0),$$

with (18) for $r = 0$ and with the unique unrestricted maximum at $t = A/r$ when $r \ge 1$. Applying this identity separately to the two terms on the right of (36) gives

$$\sup_{0<t\le t_0} \sup_{0\le\theta\le R} |q_t^{(m)}(\theta)| \le \widehat{Q}_m(t_0, R).$$

For large k, $A_k(R)/r > t_0$, so the summands reduce to a polynomial in a_k times $e^{-A_k(R)/t_0}$. Successive terms therefore admit the same kind of geometric tail as Appendix B. If $\widehat{Q}_0 < 1$, Faà di Bruno's formula and the Euler-product estimate give (23).

Cut chart

Write

$$J_{b,t}(\phi) = \frac{b^2}{t} \int_0^1 \cosh \frac{sb\phi}{t} \, \mathrm{d}s - \cosh \frac{b\phi}{t}. \tag{38}$$

Since

$$b \sinh \frac{b\phi}{t} - \phi \cosh \frac{b\phi}{t} = \phi J_{b,t}(\phi),$$

the apparent singularity at $\phi = 0$ is removed before differentiation. The heat kernel on the cut chart is exactly

$$p_t(\pi - \phi) = 2C_t \sum_{j\ge0} e^{-b_j^2/(2t)} e^{-\phi^2/(2t)} \frac{\phi}{\sin\phi} J_{b_j,t}(\phi). \tag{39}$$

We now give fully explicit derivative bounds for its three factors. The Hermite-polynomial formula implies

$$\left|\partial_\phi^r e^{-\phi^2/(2t)}\right| \le \mathcal{E}_r(t, \delta) := r! \sum_{q=0}^{\lfloor r/2 \rfloor} \frac{\delta^{r-2q}}{2^q q!(r-2q)!} \, t^{-r+q}. \tag{40}$$

Let c_n be the positive Taylor coefficients of $x/\sin x$,

$$\frac{x}{\sin x} = \sum_{n\ge0} c_n x^{2n}, \qquad c_0 = 1, \qquad c_n = \frac{(-1)^n 2(1 - 2^{2n-1}) B_{2n}}{(2n)!} \quad (n \ge 1).$$

Here B_{2n} denotes the Bernoulli numbers. For $0 < \delta < \pi$, termwise differentiation gives the convergent bound

$$\left|\partial_\phi^s \frac{\phi}{\sin\phi}\right| \le \mathcal{S}_s(\delta) := \sum_{n\ge\lceil s/2 \rceil} c_n \frac{(2n)!}{(2n-s)!} \delta^{2n-s}. \tag{41}$$

Finally, differentiation under the integral in (38) yields

$$|\partial_\phi^v J_{b,t}(\phi)| \le \mathcal{J}_v(b; t, \delta) := e^{b\delta/t} \left(\frac{b}{t}\right)^v \left(1 + \frac{b^2}{t(v+1)}\right). \tag{42}$$

Define

$$D_m^{\mathrm{cut}}(t,\delta) = 2C_t \sum_{j\geq 0} e^{-b_j^2/(2t)} \sum_{r+s+v=m} \frac{m!}{r!s!v!} \mathcal{E}_r(t,\delta)\mathcal{S}_s(\delta)\mathcal{J}_v(b_j;t,\delta). \tag{43}$$

The Gaussian $e^{-b_j^2/(2t)+b_j\delta/t}$ makes the series and all its derivatives summable. Equations (40)–(42) and the Leibniz rule prove

$$\sup_{0\leq\phi\leq\delta} |\partial_\phi^m p_t(\pi-\phi)| \leq D_m^{\mathrm{cut}}(t,\delta). \tag{44}$$

For the lower bound, the inequality used in Appendix E gives

$$J_{b,t}(\phi) = \cosh\frac{b\phi}{t}\left(\frac{b}{\phi}\tanh\frac{b\phi}{t} - 1\right) \geq \frac{b^2}{t+b\delta} - 1.$$

Under $\pi^2 > t + \pi\delta$, every summand in (39) is positive. Retaining the principal pair and using $\phi/\sin\phi \geq 1$ gives

$$p_t(\pi-\phi) \geq L_{\mathrm{cut}}(t,\delta) := 2C_t e^{-(\pi^2+\delta^2)/(2t)}\left(\frac{\pi^2}{t+\pi\delta} - 1\right) > 0. \tag{45}$$

Set

$$C_m^{\mathrm{cut}}(t,\delta) = \sum_{\ell=1}^{m} \frac{(\ell-1)!}{L_{\mathrm{cut}}(t,\delta)^\ell} \mathsf{B}_{m,\ell}\left(D_1^{\mathrm{cut}},\ldots,D_{m-\ell+1}^{\mathrm{cut}}\right). \tag{46}$$

Faà di Bruno's formula applied to $\log p_t$ proves (24). Since $\partial_\theta = -\partial_\phi$, the absolute derivative bounds are unchanged.

Antipodal asymptotics

Taking the removable limit in (39) gives the exact sum

$$p_t(\pi) = 2C_t \sum_{j\geq 0} e^{-b_j^2/(2t)}\left(\frac{b_j^2}{t} - 1\right). \tag{47}$$

The $j = 0$ term and the gap $b_1^2 - b_0^2 = 8\pi^2$ prove (26). Expanding the principal summand of (39) through order ϕ^2, then differentiating $-\log p_t$, gives

$$s_t''(\pi) = -\frac{\pi^2}{3t^2} + O(t^{-1});$$

the later pairs are smaller by $O(e^{-4\pi^2/t})$. This proves (27) and completes Theorem 6.2.

G Nonzero-image integral bound

For the negative jth image, $j \geq 1$, substitute $u = 2\pi j - \theta$ and use $|\sin\theta| \leq 1$. Extending its integration range yields a bound

$$te^{-(2j-1)^2\pi^2/(2t)}.$$

For the positive jth image the analogous bound is $te^{-2j^2\pi^2/t}$. In both sequences the ratio of consecutive bounds is at most $e^{-4\pi^2/t}$. Their sum is therefore at most $E(t)$ in (28).

For the zero image, integration by parts gives

$$\int_{\rho}^{\infty} \theta^2 e^{-\theta^2/(2t)}\,\mathrm{d}\theta = t\rho e^{-\rho^2/(2t)} + t\sqrt{\frac{\pi t}{2}}\,\operatorname{erfc}\left(\frac{\rho}{\sqrt{2t}}\right).$$

The elementary inequality $\operatorname{erfc}(z) \le e^{-z^2}/(\sqrt{\pi}z)$ gives the first term inside the brackets in (29). No denominator enclosure is needed because character orthogonality proves the normalizing denominator is exactly one.

H Rational-validation data

The JSON validation record contains:

1. the schema version and all geometric conventions;
2. the rational t_0, ρ, δ, M domain and the included image pairs;
3. tight Machin and compact Archimedean enclosures for π;
4. every Q_m, $\widehat{Q}_m$, B_m, $\widehat{B}_m$, $G_m(\rho)$, and $G_m(\pi - \delta)$, together with their omitted-tail ratios;
5. normalized tail upper bounds at the declared heat times;
6. the identity, transition, cut-pair, and plaquette positivity margins; and
7. a SHA-256 hash of the canonical JSON representation.

Every theorem-critical number is serialized as "numerator/denominator." Decimal fields are upward-rounded displays. The verifier recomputes the digest and parses the fractions itself. It reconstructs both positivity floors, the Bell-polynomial arrays, the cut-pair condition, and the displayed plaquette margins rather than accepting stored Boolean values.

The generator's elementary-function basis consists of alternating Taylor bounds, a positive-series tail bound, integer square root, Machin's identity, $e^{-x} \le 2^{-\lfloor x \rfloor}$, and $e > \sum_{k=0}^{6} 1/k! = 1957/720$. The exponential bounds are coarse by design. The SHA-256 digest establishes file integrity, not the validity of the inequalities.

I Reproducibility instructions

From the manuscript directory, generate and verify the rational record with

```
python3 supplement/exact_enclosures.py
python3 supplement/verify_certificate.py
python3 -m unittest supplement.test_heat_kernel_enclosures -v
```

Only the Python standard library is required. Compile the paper with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -outdir=build main.tex
```

The release archive contains the source, bibliography, rational record, generator, checker, tests, build report, and PDF. The exact release is identified by the SHA-256 sidecar stored beside the archive. The code is provided under the MIT license; the manuscript notice reserves its text separately.

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